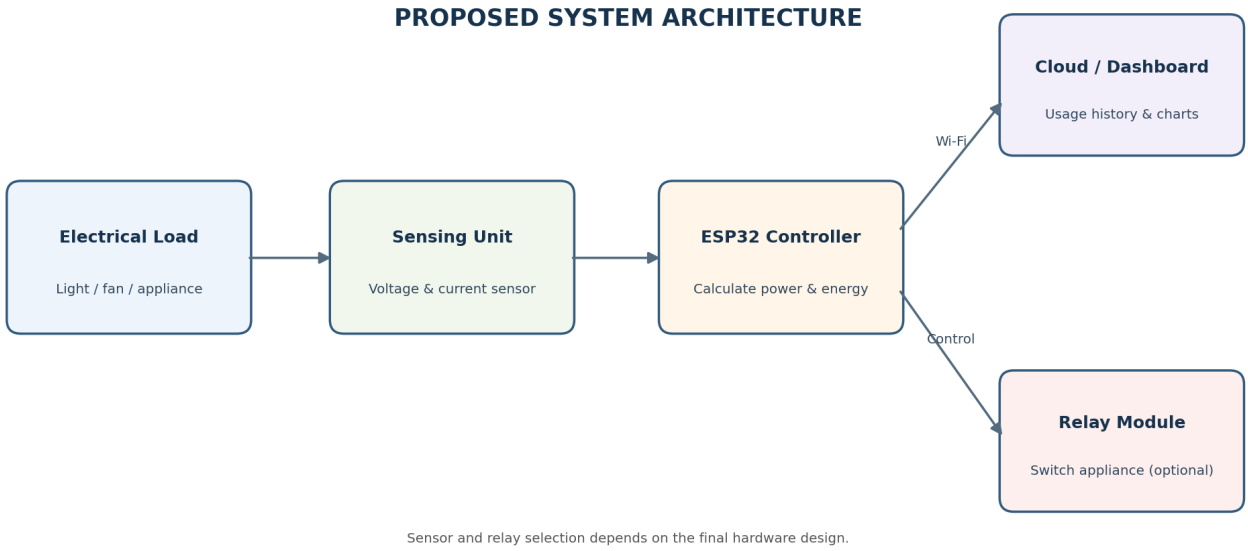
SMART HOME ENERGY MONITORING & AUTOMATION SYSTEM

Project Description, Proposed Design & Implementation Guide



Project category	Internet of Things (IoT) / Embedded Systems
Repository name	Smart-Home-Energy-Monitoring-Automation-System
Prepared for	GitHub project documentation
Project status	Proposed design — update after implementation and testing

Important: This document is a project proposal/template based on the project title only. The exact hardware, software, circuit, dashboard and results must be updated to match the system you actually build. No test results are claimed here.

1. Project Overview

The Smart Home Energy Monitoring and Automation System is an IoT-based concept for measuring household electricity usage and helping users manage connected electrical appliances. A sensing unit measures electrical parameters, a microcontroller processes the readings, and a Wi-Fi connection can send the information to a dashboard for live viewing and historical analysis. Where supported by the final design, a relay module can switch selected appliances according to user commands or predefined rules.

The main goal is to make energy consumption more visible and actionable. By understanding when and how electricity is being used, occupants can identify unnecessary consumption, make informed decisions and reduce avoidable energy waste.

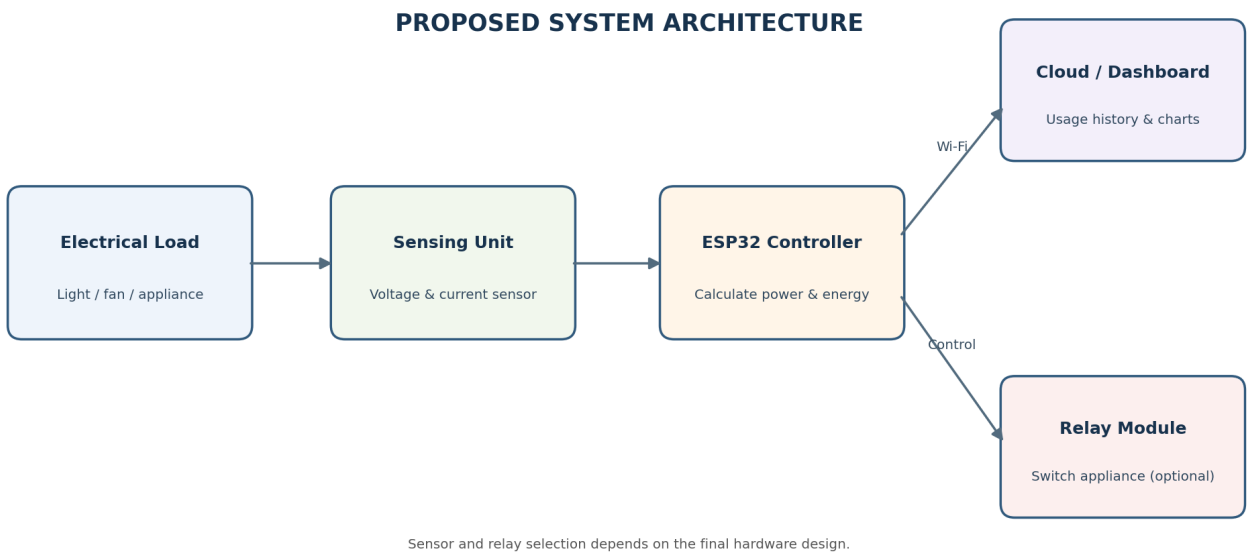
2. Problem Statement

In many homes, electricity consumption is visible only as a periodic bill. Users may not know which appliances consume the most energy or when unnecessary usage occurs. A connected monitoring system can provide more immediate information and, if automation is implemented, allow selected loads to be controlled more conveniently.

3. Objectives

- Measure electrical parameters such as voltage, current and power using an appropriate sensor.
- Estimate energy consumption over time (for example, watt-hours or kilowatt-hours).
- Display readings locally or through an IoT dashboard.
- Provide threshold-based notifications when configured limits are exceeded (optional).
- Switch selected appliances using a properly rated relay or smart switching module (optional).
- Maintain a modular design that can be expanded with additional loads or sensors.

4. Proposed System Architecture



Sensor and relay selection depends on the final hardware design.

A typical implementation consists of an electrical load, a voltage/current measurement stage, a microcontroller such as an ESP32, and a Wi-Fi-connected dashboard. An optional relay stage provides appliance switching. The final circuit must follow the selected sensor and controller datasheets.

5. Working Principle

PROJECT WORKFLOW



The sensor acquires electrical measurements. The controller samples and processes these readings, derives power and accumulated energy, and transmits selected values over Wi-Fi. The dashboard presents current readings and trends. If automation is included, the controller checks user-defined rules and operates a relay or sends an alert when a condition is met.

6. Proposed Hardware Requirements

Component	Purpose / selection note
ESP32 development board (suggested)	Main controller with Wi-Fi connectivity. Another suitable microcontroller may be used.
Energy measurement sensor (e.g., PZEM-004T-based module)	Measures electrical parameters. Confirm voltage/current range, isolation and interface compatibility.
Relay module / rated contactor (optional)	Switches a selected load. Must be correctly rated for the load and installation.
Power supply	Provides regulated low-voltage power to the controller and sensors.
Connecting wires, terminal blocks and enclosure	Provides secure connections and physical protection; choose suitable ratings.
Test load	A small, controlled load for initial testing under safe supervision.

7. Proposed Software / Tools

- Arduino IDE or PlatformIO for embedded firmware
- C/C++ for microcontroller programming
- MQTT or HTTP for data communication (choose one)
- A dashboard platform such as Blynk, Node-RED or a custom web dashboard
- Git and GitHub for version control and project documentation

Choose the tools that match your actual implementation. Do not list a platform as used until you have configured and tested it.

8. Functional Modules

Measurement: Read sensor values at a defined interval.
Processing: Calculate or obtain power and accumulated energy.
Communication: Connect to Wi-Fi and publish readings.
Dashboard: Present live measurements and historical trends.
Automation (optional): Apply thresholds or user commands to control selected loads.

9. Implementation Plan

1. Confirm the project scope, electrical ratings and required measurements.
2. Select the controller and compatible energy-measurement sensor.
3. Develop and test the low-voltage controller and sensor interface.
4. Read sensor values and verify them against a suitable reference meter.
5. Add Wi-Fi communication and display readings on the selected dashboard.
6. If required, integrate a properly rated switching stage and test its logic without exposing users to live wiring.
7. Record test cases, screenshots, limitations and measured results.
8. Publish source code, wiring diagrams, bill of materials and final report on GitHub.

10. Testing and Evaluation

After building the system, test each function separately and document the actual observations. Suggested tests include:

Test	What to verify	Evidence to add
Sensor reading	Values update consistently and within the sensor's specified range.	Serial monitor / dashboard screenshot
Measurement accuracy	Compare readings with an appropriate reference instrument under controlled conditions.	Comparison table with error calculation
Wi-Fi communication	Readings reach the dashboard and reconnect after a network interruption.	Connection log / dashboard screenshot
Energy calculation	Accumulated energy increases consistently with the measured load and time.	Recorded readings and calculation
Automation (if implemented)	Relay responds to the configured command or threshold.	Test log and safe demonstration photo

Do not invent results. Replace this section with real measurements, accuracy calculations, photos and observed limitations after testing.

11. Expected Outcomes

When implemented and validated, the project is expected to provide a clearer view of household energy consumption, make selected readings accessible through a connected interface, and support more informed energy-use decisions. Appliance automation may add convenience and help users respond to configured conditions. Actual savings depend on usage patterns and the final design.

12. Limitations and Future Enhancements

Possible limitations: measurement accuracy depends on sensor calibration and installation; Wi-Fi features depend on network availability; the design monitors only the circuits or appliances connected to its sensors; and relay control is limited by the switching device rating.

Future enhancements: multi-appliance monitoring, energy-cost estimation using a configurable tariff, data logging, mobile notifications, usage analytics, solar generation monitoring and role-based dashboard access.

13. Safety Considerations

Mains electricity can cause fatal electric shock and fire. Do not connect exposed mains wiring on a breadboard or handle live conductors. Use certified, isolated measurement equipment and correctly rated enclosures, fuses and switching devices. Have mains-side wiring, installation and testing performed or supervised by a qualified electrician. Start development with low-voltage circuits and simulated data wherever possible.

14. Suggested GitHub Repository Structure

```
Smart-Home-Energy-Monitoring-Automation-System/
■■■ README.md
■■■ docs/
■ ■■■ Project_Description.pdf
■ ■■■ Project_Report.pdf
■ ■■■ Presentation.pptx
■■■ firmware/
■■■ hardware/
■ ■■■ circuit_diagram/
■ ■■■ components_list/
■■■ dashboard/
■■■ images/
■■■ test_results/
```

15. Short GitHub Repository Summary

An IoT-based smart home system for monitoring real-time electricity consumption and managing connected appliances. It uses electrical sensors, a microcontroller and wireless communication to visualize energy usage and optionally automate loads, helping users make informed decisions and reduce energy wastage.

16. Information to Finalize Before Submission

Update this document with your actual controller, sensor model, communication protocol, dashboard, circuit diagram, source-code link, photographs, test readings, measured accuracy and known limitations. Add your name, department, institution and project guide if required by your college.